1.Data Ingestion Process

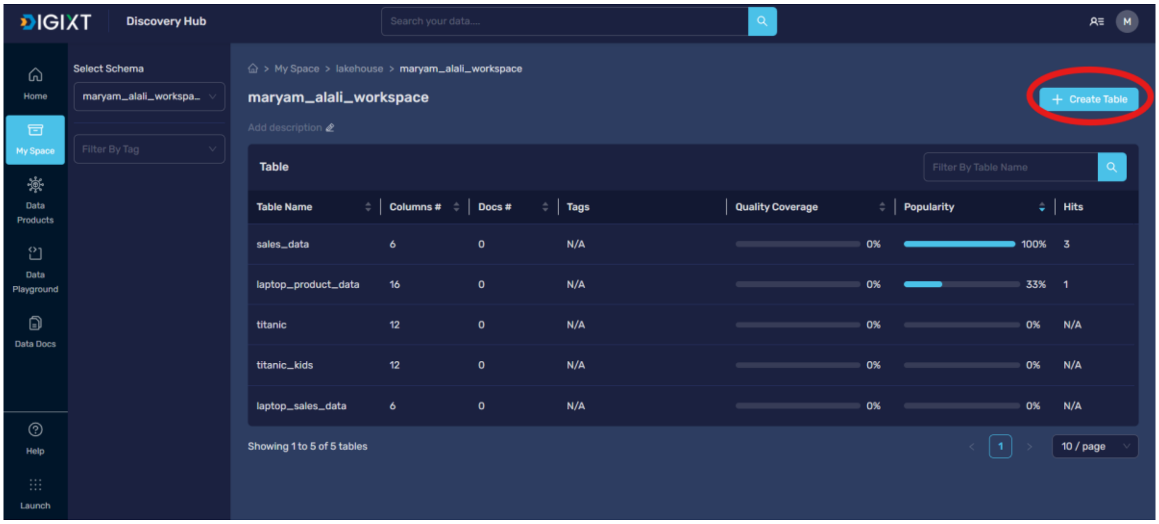

Figure 1: Ingest the data into DigiXT

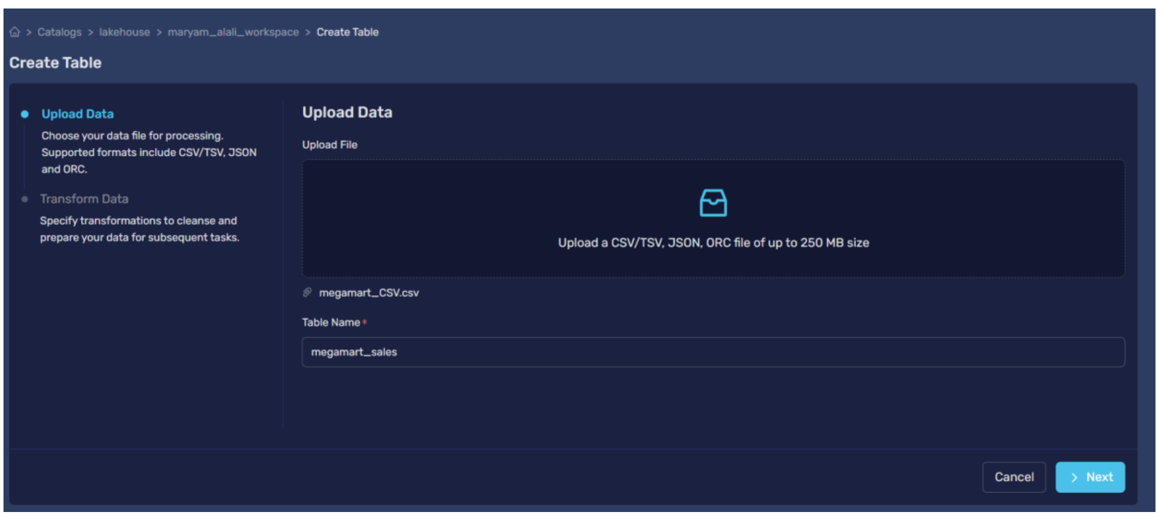

Figure 2: Upload data

The dataset was successfully loaded into the DigiXT platform. The process confirmed that all records were transferred correctly, with numerical columns like sales and profit

2 Initial Test Suite Results (Before Cleaning)

We ran a series of automated tests to check the data for Completeness, Consistency, Validity, and Accuracy

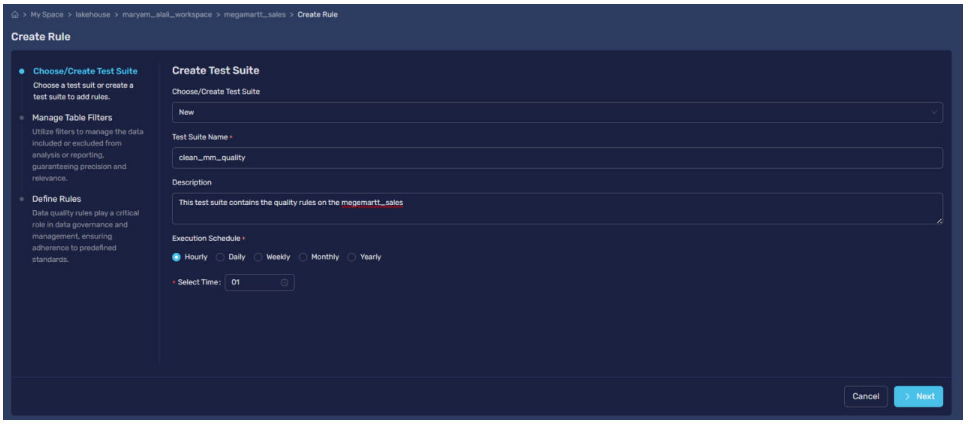

Figure 3: Create test suite

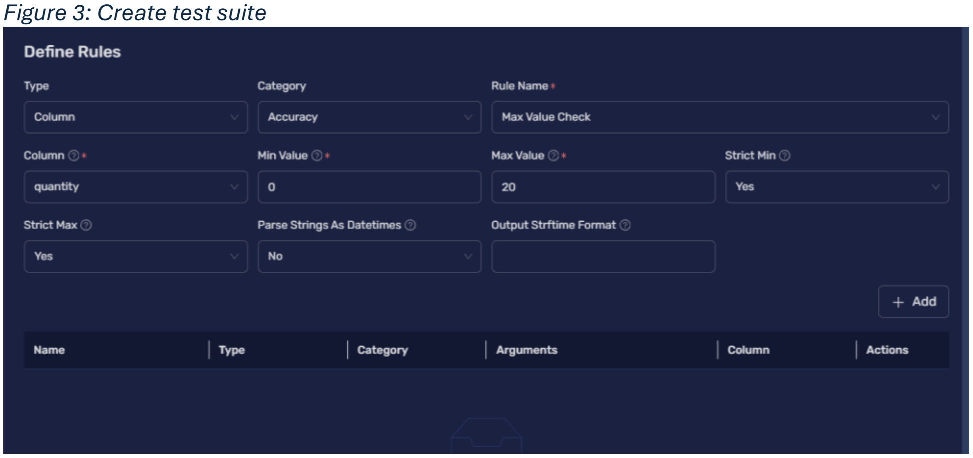

Figure 4: Rule 1 Max value check

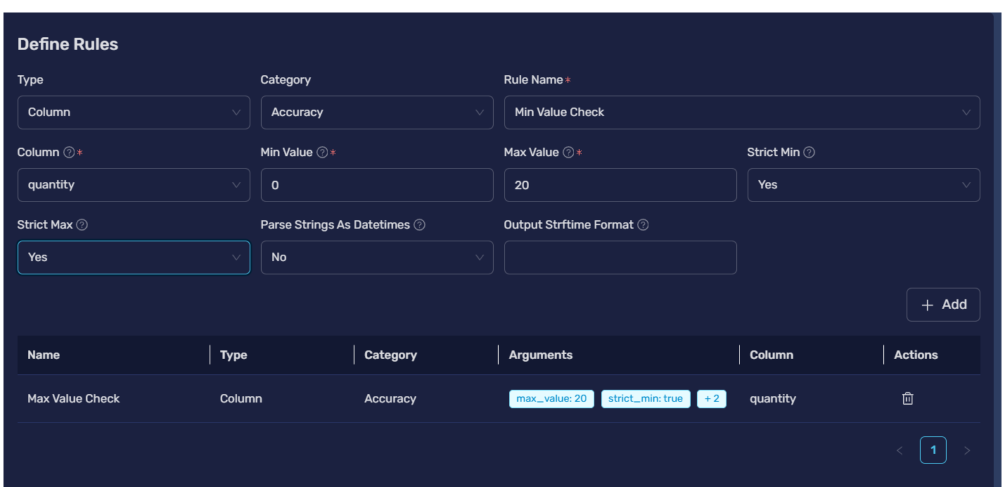

Figure 5: Rule 2 Min value check

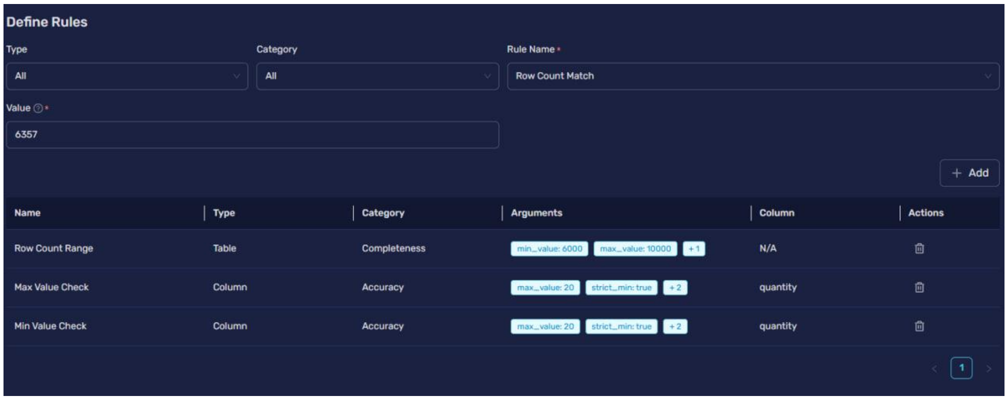

Figure 6: Rule 3 Row count match

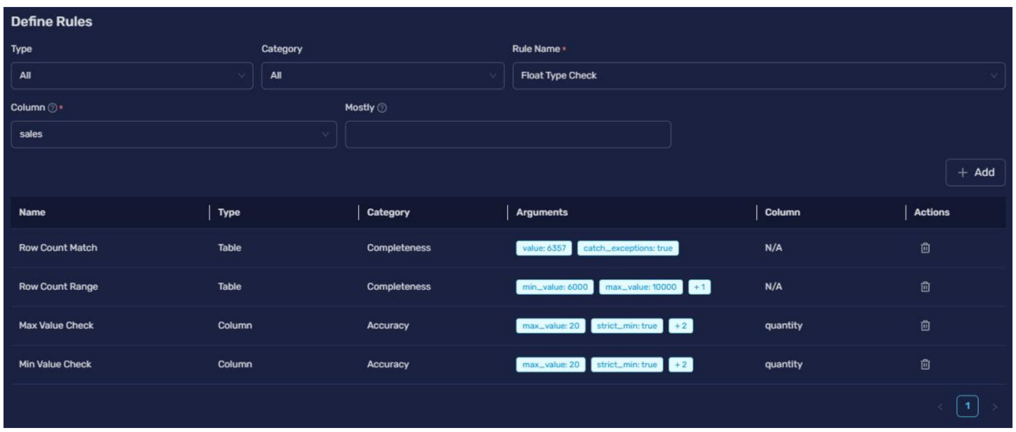

Figure 7: Rule 4 Float type check

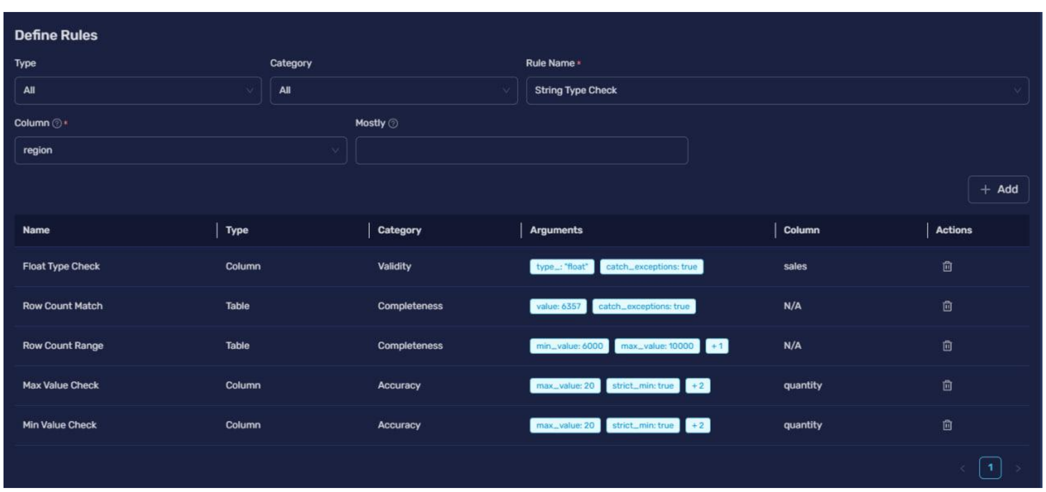

Figure 8: Rule 5 String type check

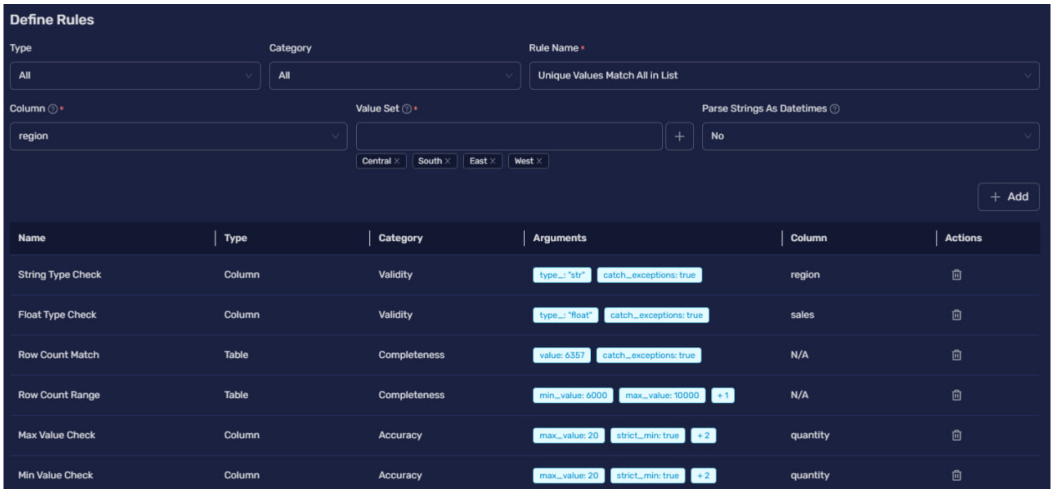

Figure 9: Rule 6 Unique Values Match All in list

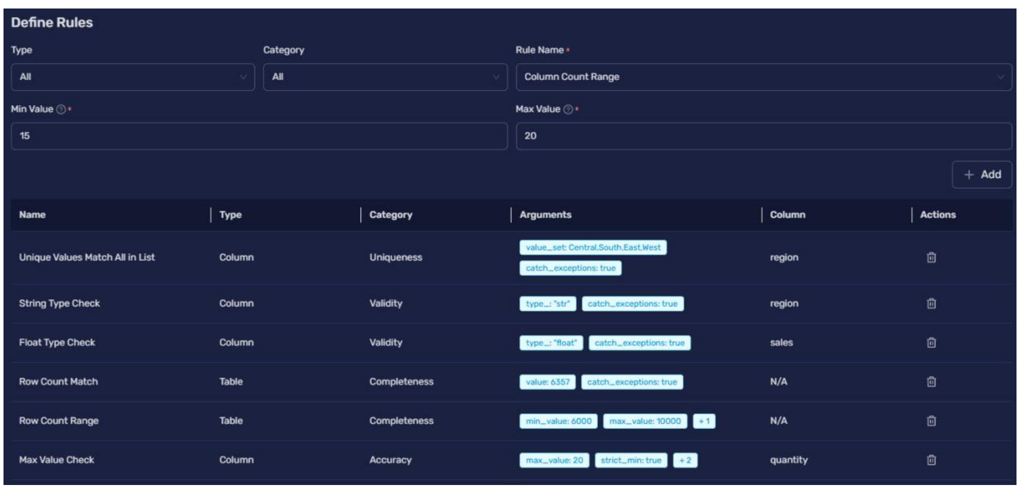

Figure 10: Rule 7 Column Count Range

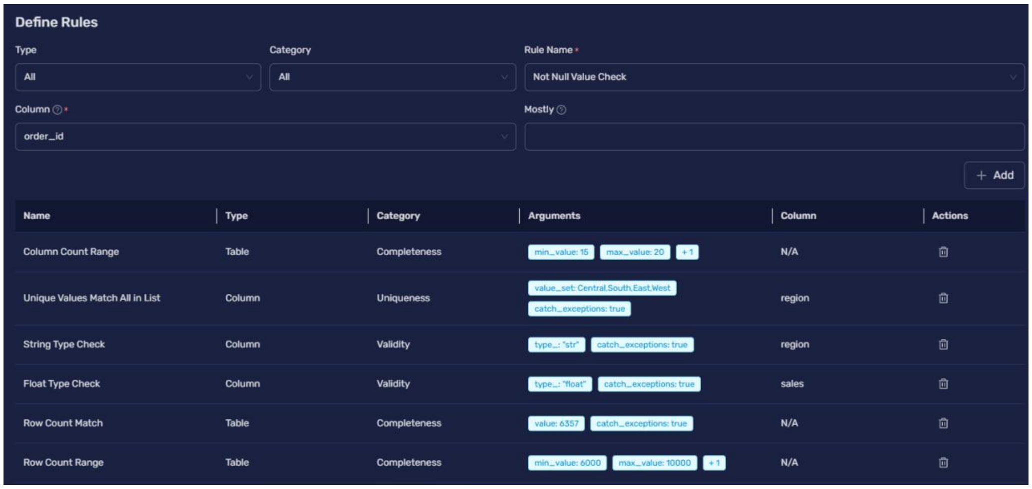

Figure 11: Rule 8 Not Null Value Check

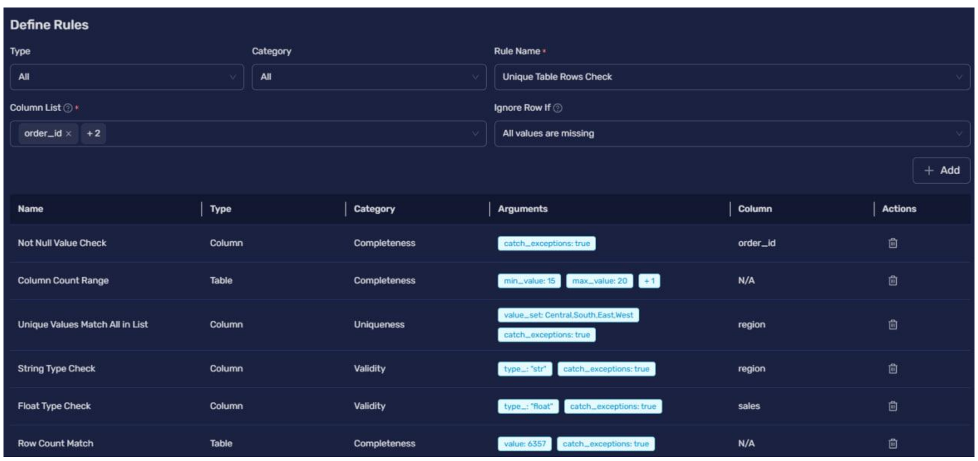

Figure 12: Rule 9 Unique Table Rows Check

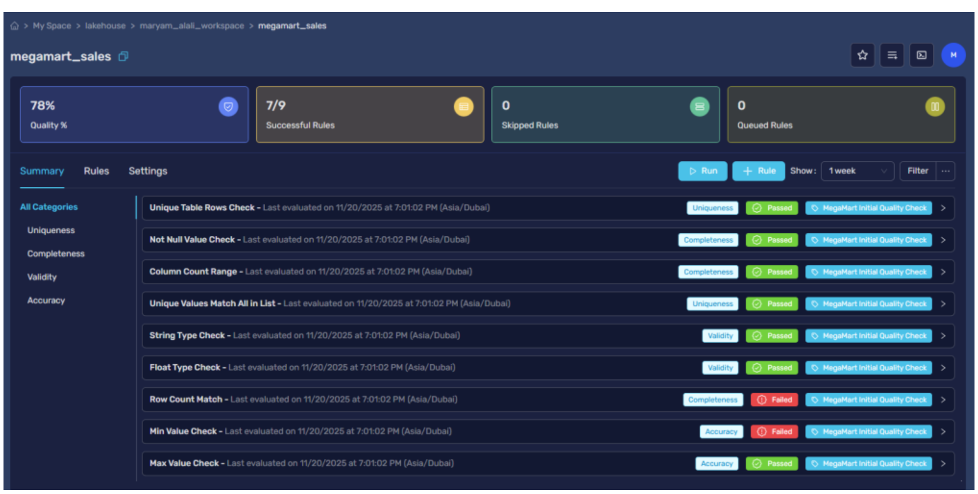

Figure 13: Data quality check before cleaning

3. Data Preprocessing Strategy

3.1. Preprocessing Objective and Methodology

By comparing the raw data file (megamart_CSV.csv) with the cleaned data file (cleaned_megamart.csv), we identified and documented the exact cleaning and preprocessing steps that were applied.

3.2. Identified Cleaning Steps and Rationale

Our comparison showed that the main data quality issues found in Task 4 were solved in the cleaned dataset, but by deleting records instead of filling in missing values. The table below documents the actions taken and the logic behind them:

Quality Issue Identified (in Raw Data)	Actions taken	Rationale & Advanced Technique
Missing Values in Key Columns (order_date, quantity, review)	Record Deletion	All records that had any missing value in these three columns were deleted.
Negative Quantity (9 records)	Record Deletion	The 9 records with negative quantities were deleted. They were likely part of the records that had missing values.
Extreme Outliers (profit)	No Action Taken (Preservation)	The outliers in the profit column were kept. Technique: Strategic Preservation, recognizing that these are not errors but real business events (like large deals or losses) that are important for a realistic analysis.

4. Re-conducting the Quality Check (Post-Curation)

To check if the applied changes were effective, we performed a final quality check on the cleaned_megamart.csv file.

Validation Check	Raw Data State	Clean Data State	Status
Total Records	9,994	6358	**VALIDATED:** 3,637 records were deleted. This solves the missing value problem.
Missing order_date	2,278	0	**VALIDATED:** All missing dates were deleted.
Missing quantity	1,257	0	**VALIDATED:** All missing quantities were deleted.
Negative quantity Count	9	0	**VALIDATED:** All negative values were deleted.

Conclusion: The change from the raw to the clean dataset significantly improved data quality in terms of completeness and validity.

The cleaned data was provided and carefully cleaned by the course instuctor

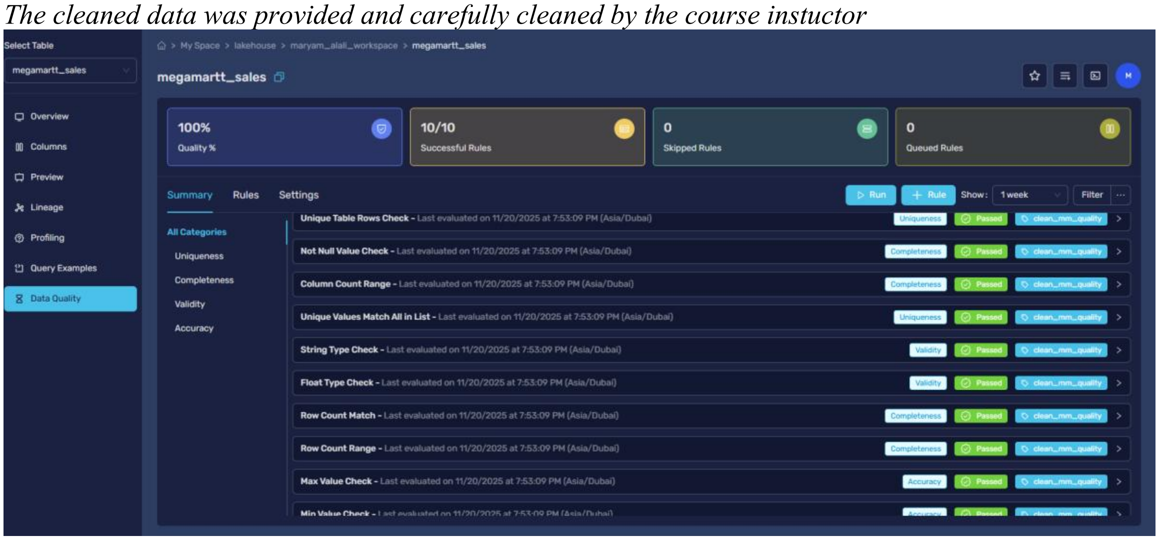

Figure 14: Data quality check after cleaning